LINUX SERVER MONITORING & TROUBLESHOOTING LAB

Step 1: Environment Setup:

Make sure your CentOS system is **up-to-date** with the latest packages and security patches

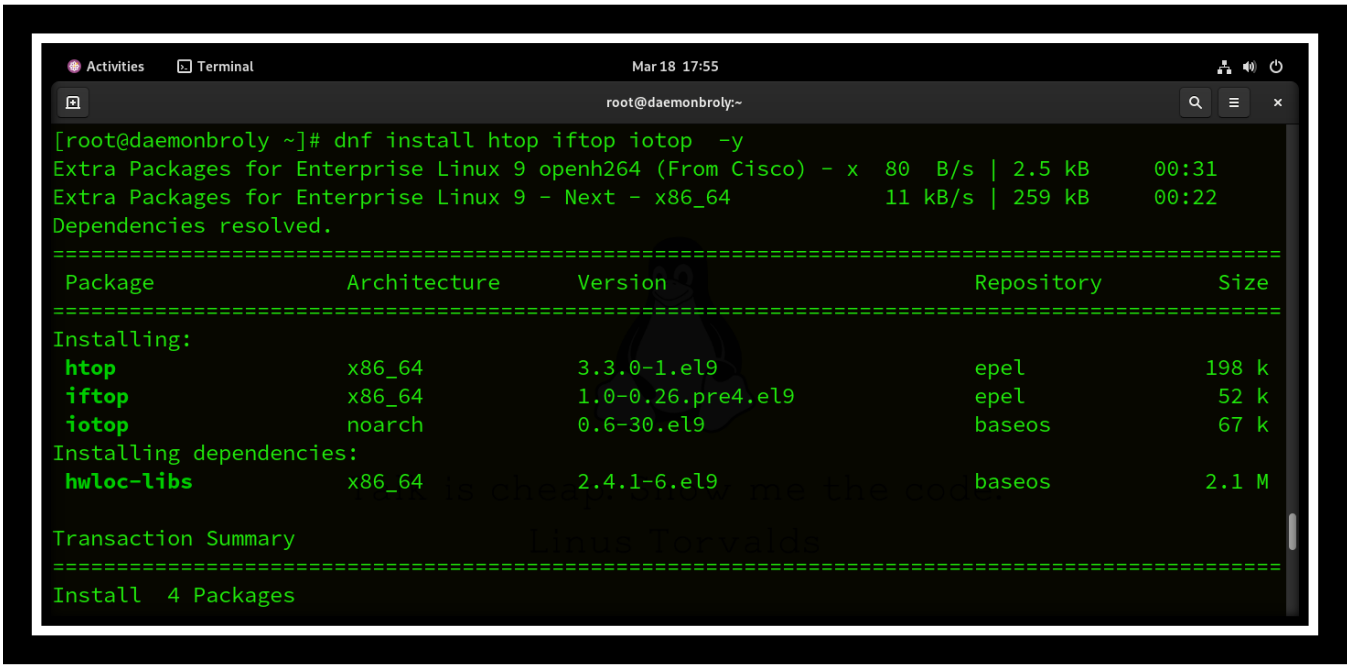
Sudo dnf update –y && upgrade –y

Enable EPEL Repository:

Sudo dnf install epel-release –y

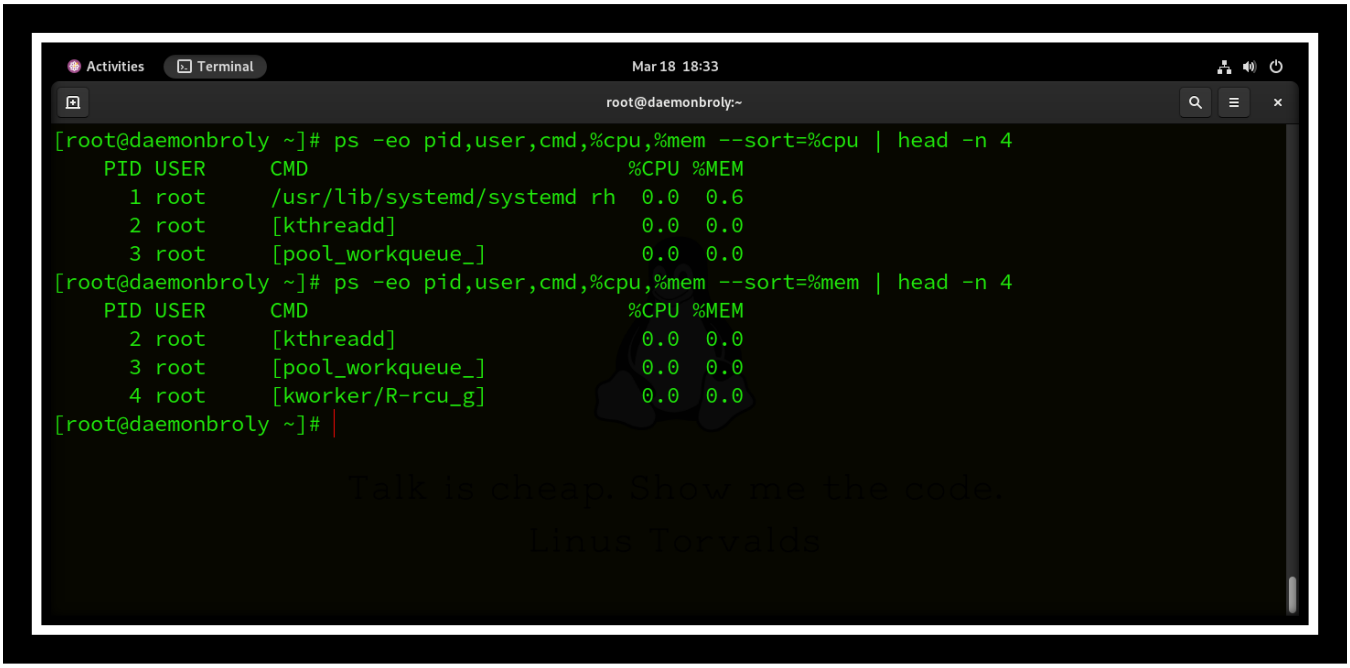
Step2:Intsall tools

Dnf install htop iftop iotop net-tools -y



Verification for download:

System Resource Monitoring



Analysing disk usage and disk free

```
[arik@daemonbroly ~]$ df -h  | sort -hrk 2 | head -n 4
/dev/mapper/cs-root   17G  7.7G  9.3G  46% /
/dev/nvme0n1p1       960M  494M  467M  52% /boot
tmpfs                908M     0  908M   0% /dev/shm
tmpfs                364M  7.5M  356M   3% /run
[arik@daemonbroly ~]$ du -h /* 2>>/dev/null | sort -hr | head -n 4
4.5G    /usr
2.0G    /var
1.7G    /var/cache
1.6G    /var/cache/PackageKit/9
[arik@daemonbroly ~]$ vmstat 2 5 | sort -hr | head -n 4
 5  0 443392 198916      0 848484    5   22   347    37   82   77  1  2 97  0  0
 1  0 443392 198916      0 848484    0    0     0     0  285  313  1  1 98  0  0
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
procs -----------memory---------- ---swap-- -----io---- -system-- ------cpu-----
[arik@daemonbroly ~]$
```

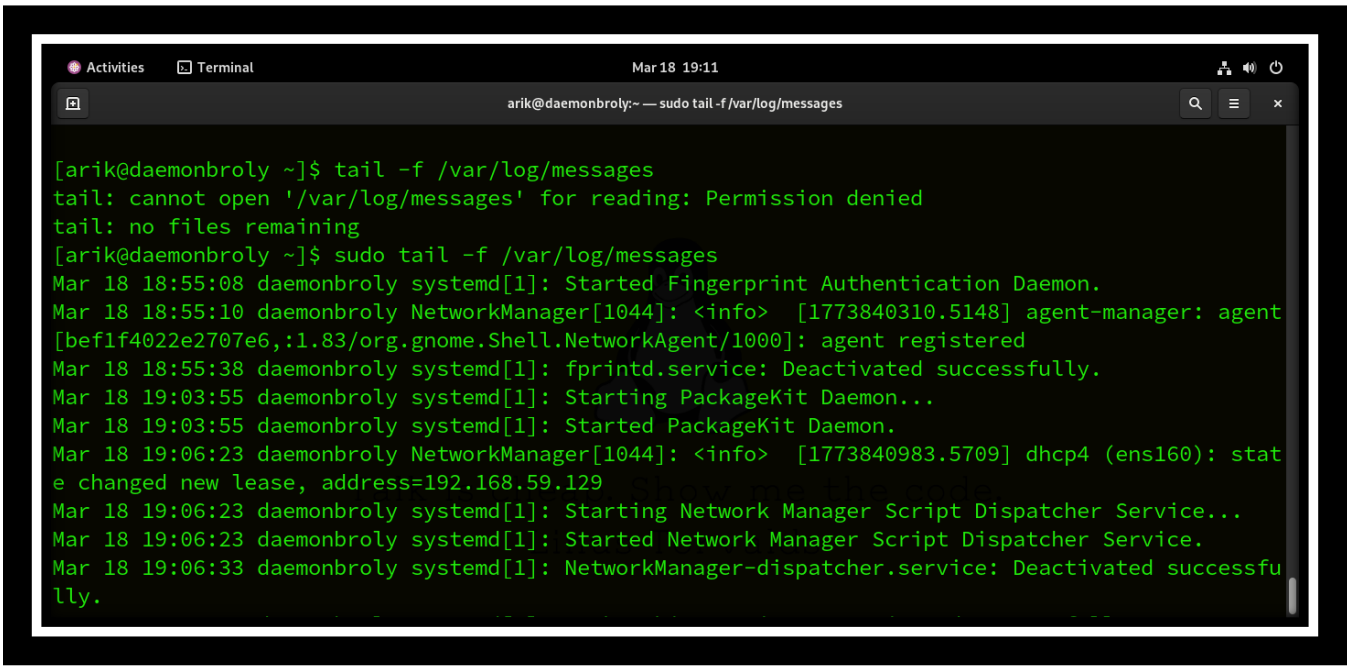
Process and Service Monitoring

```
[arik@daemonbroly ~]$ ps aux | sort -hr | head -n 5
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
rtkit        838  0.0  0.1 154016  2916 ?        SNsl 17:30   0:00 /usr/libexec/rtkit-daemon
rpcuser     1205  0.0  0.0  11356  1336 ?        Ss   17:30   0:00 /usr/sbin/rpc.statd
rpc          796  0.0  0.2  14760  4040 ?        Ss   17:30   0:00 /usr/bin/rpcbind -w -f
root          99  0.0  0.0      0     0 ?        S    17:29   0:00 [irq/29-pciehp]
[arik@daemonbroly ~]$ systemctl list-units --type=service --state=running | head -n 4
  UNIT                      LOAD   ACTIVE SUB     DESCRIPTION
  accounts-daemon.service   loaded active running Accounts Service
  alsa-state.service        loaded active running Manage Sound Card State (restore and store)
  atd.service               loaded active running Deferred execution scheduler
[arik@daemonbroly ~]$ systemctl status sshd
● sshd.service - OpenSSH server daemon
     Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-03-18 17:30:08 IST; 1h 34min ago
       Docs: man:sshd(8)
```

Process terminating:

```
[arik@daemonbroly ~]$ sleep 1000&
[1] 4588
[arik@daemonbroly ~]$ jobs
[1]+  Running                 sleep 1000 &
[arik@daemonbroly ~]$ ps
    PID TTY          TIME CMD
   4279 pts/0    00:00:00 bash
   4588 pts/0    00:00:00 sleep
   4597 pts/0    00:00:00 ps
[arik@daemonbroly ~]$ kill -9 4588
[1]+  Killed                  sleep 1000
[arik@daemonbroly ~]$
```

Network Monitoring:

```
[arik@daemonbroly ~]$ ss -tulpn | sort | head -n 4
Netid State  Recv-Q Send-Q  Local Address:Port  Peer Address:PortProcess
tcp   LISTEN 0      128           0.0.0.0:22         0.0.0.0:*
tcp   LISTEN 0      128              [::]:22            [::]:*
tcp   LISTEN 0      4096          0.0.0.0:111        0.0.0.0:*
[arik@daemonbroly ~]$ ss -t | sort | head -n 4
State Recv-Q Send-Q Local Address:Port Peer Address:Port
[arik@daemonbroly ~]$ ss -tu | sort | head -n 4
Netid State Recv-Q Send-Q         Local Address:Port      Peer Address:Port
udp   ESTAB 0      0      192.168.59.129%ens160:bootpc 192.168.59.254:bootps
[arik@daemonbroly ~]$ ss -tul | sort | head -n 4
Netid State  Recv-Q Send-Q  Local Address:Port         Peer Address:Port
tcp   LISTEN 0      128           0.0.0.0:ssh                0.0.0.0:*
tcp   LISTEN 0      128              [::]:ssh                   [::]:*
tcp   LISTEN 0      4096          0.0.0.0:40911              0.0.0.0:*
[arik@daemonbroly ~]$
```

Log Analysis & Troubleshooting